

UAV DESIGN LAB

VI-Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AE35	PCC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100
LIST OF EXPERIMENTS								
Drones: - Design calculation of frame using python code - Components Weight estimation - Power plant thrust calculation - Drone Assembly - Navigation systems Fixed wing UAV: - Wing sizing using python - Tail sizing using python - Fuselage sizing - Control surface sizing - Navigation systems Note: Ten experiments should be performed								
Reference Books: - Paul Fahlstrom, Thomas Gleason - Introduction to UAV Systems-Wiley (2012) - Reg Austin, "Unmanned Air Systems: UAV Design, Development and Deployment", First Edition, Wiley Publishers, 2015								
Software's Required Python /								
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO - Apply Python programming to perform design calculations for drone frames - Calculate power plant thrust requirements for drones, incorporating factors such as propulsion systems, weight, and aerodynamics, ensuring efficient flight performance. - Assemble and integrate drone components systematically, ensuring proper functionality and structural integrity in the final drone assembly - Employ Python programming techniques to size tails for fixed-wing UAVs, considering stability, control, and aerodynamic balance in design calculations. - Utilize Python-based methodologies to size wings for fixed-wing UAVs, incorporating aerodynamic principles and structural requirements for optimal performance.								